

Theoretical Quantum Optics: Problem Sheet 4

(for 26.05.2025)

10. Mechanical Schrödinger cats (2P)

In the setup shown in the figure below, a magnetized vibrating tip is placed on top of an isolated electronic spin qubit (associated, for example, with a NV-defect center in diamond) with states $|0\rangle$ and $|1\rangle$. The motion of the tip can be quantized and described as a harmonic oscillator with annihilation (creation) operator b ($b^\dagger$) and a vibrational frequency ω_m . When the tip moves up and down, the strong magnetic field gradient G_m generates a varying Zeeman splitting of the spin states, proportional to the displacement $z = z_0(b + b^\dagger)$. This type of coupling can be modelled by a Hamiltonian of the form

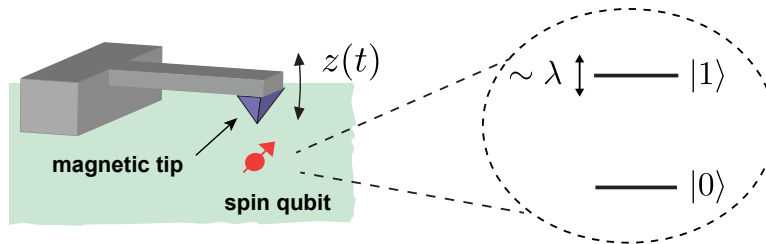
$$H = \hbar\omega_m b^\dagger b - \hbar\lambda(b + b^\dagger)|1\rangle\langle 1| \quad (1)$$

with a coupling constant $\lambda \approx z_0 G_m \mu_B / \hbar$, where μ_B is the Bohr magneton. Note that here we have already eliminated the static splitting of the spin states by moving to a rotating frame.

Let us assume that the spin is initially prepared in the superposition state $|\psi_s\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, by applying a $\pi/2$ -rotation with a classical driving field. The mechanical resonator is prepared in a coherent state $|\alpha_0\rangle$, such that $|\Psi\rangle(t=0) = |\psi_s\rangle|\alpha_0\rangle$.

- (a) Evaluate the time evolution of the combined state of this system, $|\Psi\rangle(t)$.
- (b) After the evolution time t , another fast $\pi/2$ -rotation is applied to the spin. Evaluate the probability $P_1(t)$ to find the spin in state $|1\rangle$ after this second $\pi/2$ -pulse. Plot and discuss the result for the case $\alpha_0 = 0$.
- (c) Finally, evaluate the probability $P_1(t)$ for the case where the mechanical mode is initially in a thermal state with average occupation number $\langle a^\dagger a \rangle = N_{\text{th}}$. Can one still observe revivals of $P_1(t)$?

Hints: For this exercise, follow the ideas discussed in Sec. 3.3.2 and use the results for the driven harmonic oscillator in Sec. 2.3.2. For the last point, evaluate $P_\uparrow(t)$ for a given α_0 and then perform an average over the thermal distribution $P(\alpha_0) = e^{-\frac{|\alpha_0|^2}{N_{\text{th}}}} / (\pi N_{\text{th}})$.



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(a) Evaluate the time evolution of the combined state of this system, $|\Psi\rangle(t)$.

(b) After the evolution time t , another fast $\pi/2$ -rotation is applied to the spin. Evaluate the probability

$$|z_0\rangle = |z_s\rangle|\alpha_0\rangle = \frac{1}{\sqrt{2}}(|0\alpha_0\rangle + |1\alpha_0\rangle)$$

$$|u(t)\rangle = e^{-iHt} |z_0\rangle$$

$$e^{-iHt} = e^{-i\omega_m b^\dagger b t} e^{i\lambda(b + b^\dagger) t \sigma_x}$$

$$e^{i\lambda(b + b^\dagger)t}$$

$$\hat{b}^\dagger b, \hat{b} + \hat{b}^\dagger$$

$$A = \hat{b}^\dagger \hat{b}$$

$$B = (\hat{b} + \hat{b}^\dagger)$$

$$[A, [B, A]] = -[\hat{b}^\dagger \hat{b}, [\hat{b}^\dagger \hat{b}, [\hat{b} + \hat{b}^\dagger]]]$$

$$= -[\hat{b}^\dagger \hat{b}, \{[\hat{b}^\dagger \hat{b}, \hat{b}] + [\hat{b}^\dagger \hat{b}, \hat{b}^\dagger]\}]$$

$$= -[\hat{b}^\dagger \hat{b}, \{[\hat{b}^\dagger \hat{b}, \hat{b}] + [\hat{b}^\dagger \hat{b}, \hat{b}^\dagger]\}]$$

$$= -[\hat{b}^\dagger \hat{b}, -\hat{b} + \hat{b}^\dagger]$$

$$= +[\hat{b}^\dagger \hat{b}, \hat{b} - \hat{b}^\dagger]$$

$$= [\hat{b}^\dagger, \hat{b}]\hat{b} - \hat{b}^\dagger[\hat{b}, \hat{b}^\dagger]$$

$$= -\hat{b} - \hat{b}^\dagger$$

$$[B, [B, A]] = [\hat{b} + \hat{b}^\dagger, [\hat{b}^\dagger \hat{b}, [\hat{b} + \hat{b}^\dagger]]] = [\hat{b} + \hat{b}^\dagger, \{[\hat{b}^\dagger \hat{b}, \hat{b}] + [\hat{b}^\dagger \hat{b}, \hat{b}^\dagger]\}]$$

$$= [\hat{b} + \hat{b}^\dagger, -\hat{b} + \hat{b}^\dagger] = [\hat{b}^\dagger \hat{b}, \hat{b}] - [\hat{b}^\dagger \hat{b}, \hat{b}^\dagger]$$

$$= -\hat{b} - \hat{b}^\dagger = \hat{b} + \hat{b}^\dagger$$

force or driving field:

Hamiltonian: $H(t) = \hbar\omega_c a^\dagger a - \hbar f(t)(a + a^\dagger)$

free evolution + linear d force

no subject to some external force

arbitrary time-dependent force

• time evolution:

end up in a coherent state under this force

$$|\psi\rangle(0) = |\alpha_0\rangle \Rightarrow |\psi\rangle(t) = e^{i\phi(t)}|\alpha(t)\rangle$$

initial coherent state, including vacuum state

$$\alpha(t) = e^{-i\omega_c t} \alpha_0 + i \int_0^t ds f(s) e^{-i\omega_c(t-s)}$$

$$\phi(t) = \int_0^t ds_1 \int_0^{s_1} ds_2 f(s_1) f(s_2) \sin(\omega_c(s_1 - s_2))$$

$$\psi(x) = \int_0^t ds_1 \int_0^{s_1} ds_2 \lambda^2 |x| \sin(\omega_m(s_1 - s_2))$$

$$a|\alpha\rangle = \alpha|\alpha\rangle \Rightarrow |\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle,$$

$$\rightarrow e^{-iHt/\hbar} |0\rangle = e^{-i\omega_m n t - 0} = e^{-i\hbar m n t} |0\rangle$$

$$e^{-iHt/\hbar} |1\rangle = e^{-i\hbar m n} |1\rangle$$

$$e^{i\hat{b}t} |\alpha\rangle = e^{i\alpha t} |\alpha\rangle$$

$$e^{i\hat{b}^\dagger t} |\alpha\rangle$$

$$e^{i(\hbar\omega_m \hat{b}^\dagger \hat{b} - (\hat{b} + \hat{b}^\dagger)\lambda |x|)t} e^{\hat{b}^\dagger - \alpha \hat{b}}$$

$$\lambda \hat{b} |x| + \hat{b}^\dagger \lambda |x|$$

$$H = \hbar\omega_m b^\dagger b - \hbar\lambda(b + b^\dagger)|1\rangle\langle 1|$$

$$\rightarrow H = \frac{\hbar\omega_m}{2} \left[(b + b^\dagger)^2 - 1 - b^{\dagger 2} - b^2 \right] - \hbar\lambda(b + b^\dagger)$$

$$b\hat{b} = 1 + \hat{b}^\dagger \hat{b} = \frac{\omega_m}{2} (\hat{b}^\dagger + \hat{b})^2 - \lambda(\hat{b}^\dagger \hat{b}) - \frac{\omega_m}{2} \hat{b}^2 - \frac{\omega_m}{2} \hat{b}^{\dagger 2} - \frac{\omega_m}{2} 1$$

$$(\hat{b} + \hat{b}^\dagger)^2 = (\hat{b} + \hat{b}^\dagger)(\hat{b} + \hat{b}^\dagger) = \hat{b}^2 + \hat{b}^{\dagger 2} + \hat{b}\hat{b}^\dagger + \hat{b}^\dagger\hat{b} = \hat{b}^2 + \hat{b}^{\dagger 2} + 2\hat{b}\hat{b}^\dagger + 1 = \frac{\omega_m}{2} x^2 - \lambda x + \frac{\lambda^2}{2\omega_m} - \frac{\lambda^2}{2\omega_m} - \frac{\omega_m}{2} \hat{b}^2 + \hat{b}^{\dagger 2} - \frac{\omega_m}{2} 1$$

$$\hat{b}\hat{b} = \frac{(\hat{b} + \hat{b}^\dagger)^2 - 1 - \hat{b}^2 - \hat{b}^{\dagger 2}}{2}$$

$$\alpha = \sqrt{\frac{\omega_m}{2}} x$$

$$2\alpha b = \lambda = 2\sqrt{\frac{\omega_m}{2}} b$$

$$\Rightarrow b = \frac{\lambda}{2} \sqrt{\frac{2}{\omega_m}} = \frac{\lambda}{\sqrt{2\omega_m}}$$

$$\frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$= \frac{\omega_m}{2} x^2 - \lambda x + \frac{\lambda^2}{2\omega_m} - \frac{\lambda^2}{2\omega_m} - \frac{\omega_m}{2} \hat{b}^2 + \hat{b}^{\dagger 2} - \frac{\omega_m}{2} 1$$

$$= \left(\frac{\omega_m}{2} (\hat{b} + \hat{b}^\dagger)^2 - \frac{\lambda^2}{2\omega_m} \right) - \frac{\lambda}{\sqrt{2\omega_m}} - \frac{\omega_m}{2} (\hat{b}^2 + \hat{b}^{\dagger 2} - 1)$$

$$[\hat{b}^\dagger, \hat{b}^\dagger, \hat{b}^\dagger \hat{b}^{\dagger 2}] = [\hat{b}^\dagger, \hat{b}^2] + [\hat{b}^\dagger, \hat{b}^{\dagger 2}] = [\hat{b}^\dagger, \hat{b}] \hat{b} + [\hat{b}^\dagger, \hat{b}^2] \hat{b}^{\dagger 2}$$

here:

$$H = \hbar\omega_m b^\dagger b - \hbar\lambda(b + b^\dagger)|1\rangle\langle 1|$$

$$f(t) = \lambda |1\rangle\langle 1|$$

$$\alpha(t) = e^{-i\omega_m t} \alpha_0 + i \int_0^t ds \lambda |1\rangle\langle 1| e^{-i\omega_m(t-s)}$$

$$= e^{-i\omega_m t} \alpha_0 + i \lambda |1\rangle\langle 1| \int_0^t ds e^{-i\omega_m(t-s)}$$

$$= e^{-i\omega_m t} \alpha_0 + \frac{i\lambda}{2} |1\rangle\langle 1|$$

$$|\alpha\rangle = \mathcal{D}(\alpha)|0\rangle$$

$$\mathcal{D}(\alpha) = e^{\alpha a^\dagger - \alpha^* a}$$

Ex 1a)

$$| \psi_0 \rangle = | \psi_0 \rangle | \alpha_0 \rangle = \frac{1}{\sqrt{2}} (| 0 \alpha_0 \rangle + | 1 \alpha_0 \rangle)$$

$$e^{-i H t} = e^{-i \omega_m b^\dagger b t + i \lambda (b + b^\dagger) \tau \sigma_x}$$

$$U(t) | 0 \alpha_0 \rangle = e^{-i \omega_m b^\dagger b t} | 0 \alpha_0 \rangle = | e^{-i \omega_m t} \alpha_0 \rangle$$

$$e^{i p b^\dagger b} | \alpha \rangle = e^{i \lambda b^\dagger b} \sum_n \frac{1}{n!} e^{-|\alpha|^2/2} (\alpha^n / n!) | n \rangle = e^{-|\alpha|^2/2} \sum_n e^{i \lambda \alpha^n / n!} \frac{1}{n!} | n \rangle = | e^{i \lambda} \alpha \rangle$$

$\uparrow$
 $| e^{i \lambda} \alpha | = \alpha$

$U(t) | 1 \alpha_0 \rangle$:

Use force or driving field:

free evolution + linear d. force

Hamiltonian: $H(t) = \hbar \omega_c a^\dagger a - \hbar f(t)(a + a^\dagger)$ arbitrary time-dependent force

$\Rightarrow$ subject to some external force

time evolution: end up in a coherent state under this force

$$| \psi \rangle(0) = | \alpha_0 \rangle \Rightarrow | \psi \rangle(t) = e^{i \phi(t)} | \alpha(t) \rangle$$

initial coherent state, including vacuum state

$$\alpha(t) = e^{-i \omega_c t} \alpha_0 + i \int_0^t ds f(s) e^{-i \omega_c (t-s)}$$

$$\phi(t) = \int_0^t ds_1 \int_0^{s_1} ds_2 f(s_1) f(s_2) \sin(\omega_c (s_1 - s_2))$$

$$\begin{aligned} \phi(t) &= \int_0^t ds_1 \int_0^{s_1} ds_2 \lambda^2 \sin(\omega_c (s_1 - s_2)) = \lambda^2 \int_0^t ds_1 \int_0^{s_1} ds_2 (\cos(\omega_c s_2) \sin(\omega_c s_1) - \sin(\omega_c s_2) \cos(\omega_c s_1)) \\ &= \lambda^2 \left(\int_0^t ds_1 \sin(\omega_c s_1) \int_0^{s_1} ds_2 \cos(\omega_c s_2) - \int_0^t ds_1 \cos(\omega_c s_1) \int_0^{s_1} ds_2 \sin(\omega_c s_2) \right) = \lambda^2 \left(\int_0^t ds_1 \sin(\omega_c s_1) \frac{1}{\omega_c} \sin(\omega_c s_1) + \int_0^t ds_1 \cos(\omega_c s_1) \frac{1}{\omega_c} (\cos(\omega_c s_1) - 1) \right) \\ &= \frac{\lambda^2}{\omega_c} \left(\int_0^t ds_1 \sin^2(\omega_c s_1) + \int_0^t ds_1 \cos^2(\omega_c s_1) \int_0^t ds_1 \cos(\omega_c s_1) \right) = \frac{\lambda^2}{\omega_c} \left(\int_0^t ds_1 (1 - \cos(\omega_c s_1)) \right) \\ &= \frac{\lambda^2}{\omega_c} \left[s_1 - \sin(\omega_c s_1) \frac{1}{\omega_c} \right]_0^t = \frac{\lambda^2}{\omega_c} (t + \sin(\omega_c t)) \\ \Rightarrow | \psi(t) \rangle &= \frac{1}{\sqrt{2}} \left(| e^{-i \omega_m t} \alpha_0 \rangle | 0 \rangle + e^{i \frac{\lambda^2}{\omega_c} (t + \sin(\omega_c t))} | 1 \rangle | e^{-i \omega_m t} \left(\alpha_0 + \frac{\lambda}{\omega_m} (e^{i \omega_m t} - 1) \right) \rangle \right) \end{aligned}$$

b) second $\frac{\pi}{2}$ pulse

$$| \psi \rangle = \frac{1}{2} \left(| e^{-i \omega_m t} \alpha_0 \rangle (| 0 \rangle + | 1 \rangle) + e^{i \frac{\lambda^2}{\omega_c} (t + \sin(\omega_c t))} (| 0 \rangle - | 1 \rangle) | e^{-i \omega_m t} \left(\alpha_0 + \frac{\lambda}{\omega_m} (e^{i \omega_m t} - 1) \right) \rangle \right)$$

$$= \frac{1}{2} \left(| 0 \rangle \left(| e^{-i \omega_m t} \alpha_0 \rangle + e^{i \frac{\lambda^2}{\omega_c} (t + \sin(\omega_c t))} | e^{-i \omega_m t} \left(\alpha_0 + \frac{\lambda}{\omega_m} (e^{i \omega_m t} - 1) \right) \rangle \right) + | 1 \rangle \left(| e^{-i \omega_m t} \alpha_0 \rangle - e^{i \frac{\lambda^2}{\omega_c} (t + \sin(\omega_c t))} | e^{-i \omega_m t} \left(\alpha_0 + \frac{\lambda}{\omega_m} (e^{i \omega_m t} - 1) \right) \rangle \right) \right)$$

$$\Rightarrow P_1(t) = \frac{1}{4} \left| | e^{-i \omega_m t} \alpha_0 \rangle - e^{i \frac{\lambda^2}{\omega_c} (t + \sin(\omega_c t))} | e^{-i \omega_m t} \left(\alpha_0 + \frac{\lambda}{\omega_m} (e^{i \omega_m t} - 1) \right) \rangle \right|^2$$

for $\alpha_0 = 0$:

$$P_n(t) = \frac{1}{4} |10\rangle - e^{i \frac{\lambda^2}{\omega} (t + \sin(\omega t))} \left| \left(e^{-i\omega t} \frac{\lambda}{\omega} (e^{+i\omega t} - 1) \right) \right\rangle \Big|^2$$

$$= \frac{1}{4} \left(\langle 0|0\rangle - e^{i\phi(t)} \langle 0| \frac{\lambda}{\omega} (1 - e^{-i\omega t}) \rangle + e^{-i\phi(t)} \langle \frac{\lambda}{\omega} (1 - e^{-i\omega t}) | 0 \rangle + \langle \frac{\lambda}{\omega} (1 - e^{-i\omega t}) | \frac{\lambda}{\omega} (1 - e^{-i\omega t}) \rangle \right)$$

$$\langle \beta | \alpha \rangle = e^{i \text{Im}\{\alpha \beta^*\}} e^{-|\alpha - \beta|^2/2}$$

$$= \frac{1}{4} (1 + \cos(\phi(t)) (e^{i\phi(t)} + e^{-i\phi(t)}) + 1) = 1$$

$$\rightarrow \langle 0 | \alpha(t) \rangle = e^0 e^{-|\alpha(t)|^2/2} = 1 \cdot e^{-|\alpha(t)|^2/2} = \exp\left(-\frac{\lambda^2}{2\omega^2} |1 - e^{-i\omega t}|^2\right)$$

$$\rightarrow \langle \alpha(t) | 0 \rangle = \overline{\langle 0 | \alpha(t) \rangle} = \langle 0 | \alpha(t) \rangle$$

$$\begin{aligned} & \left| \frac{\lambda}{\omega} (1 - e^{-i\omega t}) \right|^2 \\ &= \frac{\lambda^2}{\omega^2} |1 - e^{-i\omega t}|^2 \end{aligned}$$

$$(\#) = \frac{1}{4} (2 + 2 \cos(\phi(t)) e^{-\frac{\lambda^2}{2\omega^2} |1 - e^{-i\omega t}|^2}) = \frac{1}{2} \left(1 + \cos\left(\frac{\lambda^2}{\omega} (t + \sin(\omega t))\right) e^{-\frac{\lambda^2}{2\omega^2} |1 - e^{-i\omega t}|^2} \right)$$

11. Schriffer-Wolff transformation (2P)

Consider two TLAs that are coupled to the same cavity mode, as described by the Hamiltonian

$$H = -\hbar\Delta \sum_{i=1,2} |e\rangle_i \langle e| + \frac{\hbar}{2} \sum_{i=1,2} g_i (\sigma_+^i a + \sigma_-^i a^\dagger) \equiv H_0 + H_g. \quad (2)$$

Use a Schriffer-Wolff transformation, $\tilde{H} = e^{iS} H e^{-iS}$, to derive an effective Hamiltonian valid up to order $O(g^2)$ and show that it is given by

$$H = \sum_{i=1,2} \left[-\hbar\Delta |e\rangle_i \langle e| - \frac{\hbar g_i^2}{4\Delta} (a^\dagger a \sigma_z^i + |e\rangle_i \langle e|) \right] - \frac{\hbar g_1 g_2}{4\Delta} (\sigma_+^1 \sigma_-^2 + \sigma_-^1 \sigma_+^2). \quad (3)$$

Hint: Follow the derivation for a single TLA from the lecture.

12. Markov approximation at work (2P)

In the lecture we introduced the following toy model for atomic decay

$$H = \omega_{eg} |e\rangle \langle e| + \sum_{n=1}^M \omega_n a_n^\dagger a_n + \frac{g}{\sqrt{2\pi M}} \sum_{n=1}^M (|n\rangle \langle e| + |e\rangle \langle n|), \quad (4)$$

where the $\omega_n \in [\omega_{eg} - B, \omega_{eg} + B]$ are equally spaced within a frequency band of width $2B$.

(a) Consider the following unitary transformation

$$|\psi\rangle(t) = U(t) |\tilde{\psi}\rangle, \quad U(t) = e^{-iH_0 t}, \quad H_0 = \omega_{eg} |e\rangle \langle e| + \omega_{eg} \sum_{n=1}^M |n\rangle \langle n| \quad (5)$$

to eliminate the fast frequency scale ω_{eg} . Insert this ansatz into the Schrödinger equation and derive the Hamiltonian $\tilde{H}$ that determines the evolution of $|\tilde{\psi}\rangle(t)$.

Remark: This unitary transformation differs from the usual transformation into the interaction picture and leads to a time-independent Hamiltonian $\tilde{H}$, which is more useful for numerics.

(b) Write a Python program to simulate the dynamics of $|\tilde{\psi}(t)\rangle$ starting from an initial state $|\tilde{\psi}\rangle(t=0) = |e\rangle|\text{vac}\rangle$. Plot the evolution of the atomic population $p_e(t) = |c_e(t)|^2$ for different parameters M , B and g .

(c) Discuss the results. Under which conditions can you see an exponential decay? Why are there recurrences and at which times do they occur?

11.

$$H = -\hbar \Delta \sum_{i=1,2} |e_i\rangle\langle e_i| + \frac{\hbar}{2} \sum_{i=1,2} g_i (\hat{\sigma}_i^+ \hat{a} + \hat{\sigma}_i^- \hat{a}^\dagger) = H_0 + H_g$$

Schrieffer-Wolff:

$$H = H_0 + H_g \quad U = e^{iS} \rightarrow \tilde{H} = U H U^\dagger = H_0 + H_g + i[S, H_0] + i[S, H_g] - \frac{i^2}{2}[S, [S, H_0]] + \dots$$

$$\Rightarrow i[S, H_0] = -H_g$$

$$1 \text{ atom: } S = i \frac{g}{\omega_0} (\hat{\sigma}_i^+ \hat{a} - \hat{\sigma}_i^- \hat{a}^\dagger)$$

$$\text{guess for 2 atoms: } 2S = i \frac{g}{\omega_0} (\hat{\sigma}_1^+ \hat{a}_1 - \hat{\sigma}_1^- \hat{a}_1^\dagger) + i \frac{g}{\omega_0} (\hat{\sigma}_2^+ \hat{a}_2 - \hat{\sigma}_2^- \hat{a}_2^\dagger)$$

$$H_0 = -\hbar \Delta (|e_1\rangle\langle e_1| + |e_2\rangle\langle e_2|)$$

$$i[S, H_0] = + \frac{1}{\omega_0} \hbar \Delta \left(g_1 (\hat{\sigma}_1^+ \hat{a}_1 - \hat{\sigma}_1^- \hat{a}_1^\dagger), |e_1\rangle\langle e_1| + g_2 (\hat{\sigma}_2^+ \hat{a}_2 - \hat{\sigma}_2^- \hat{a}_2^\dagger), |e_2\rangle\langle e_2| + g_1 (\hat{\sigma}_1^+ \hat{a}_1 - \hat{\sigma}_1^- \hat{a}_1^\dagger), |e_2\rangle\langle e_2| + g_2 (\hat{\sigma}_2^+ \hat{a}_2 - \hat{\sigma}_2^- \hat{a}_2^\dagger), |e_1\rangle\langle e_1| \right)$$

$$= \hbar \Delta (g_1 [\hat{\sigma}_1^+, |e_1\rangle\langle e_1|] \hat{a}_1 - [\hat{\sigma}_1^-, |e_1\rangle\langle e_1|] \hat{a}_1^\dagger + g_2 [\hat{\sigma}_2^+, |e_2\rangle\langle e_2|] \hat{a}_2 - [\hat{\sigma}_2^-, |e_2\rangle\langle e_2|] \hat{a}_2^\dagger) \\ = \hbar \Delta (g_1 (\hat{a}_1 \hat{\sigma}_1^+ - \hat{\sigma}_1^- \hat{a}_1^\dagger) - g_2 (\hat{a}_2 \hat{\sigma}_2^+ - \hat{\sigma}_2^- \hat{a}_2^\dagger)) = 2H_g \Rightarrow \text{the guess for } S \text{ works}$$

$$[S, [S, H_0]] = \frac{i}{\omega_0} [S, i[S, H_0]] = i[S, H_g] \Rightarrow i[S, H_g]$$

$$i[S, H_g] = \frac{i}{\omega_0} [S, [S, H_0]] = i[S, H_g] = \frac{i}{\omega_0} [S, H_g]$$

$$\Rightarrow \tilde{H} = H_0 + \frac{i}{\omega_0} [S, H_g] + \mathcal{O}(g^2)$$

$$i[S, H_g] = \frac{-\hbar}{\omega_0} \left([g_1 (\hat{\sigma}_1^+ \hat{a}_1 - \hat{\sigma}_1^- \hat{a}_1^\dagger), g_1 (\hat{\sigma}_1^+ \hat{a}_1 + \hat{\sigma}_1^- \hat{a}_1^\dagger)] + [g_1 (\hat{\sigma}_1^+ \hat{a}_1 - \hat{\sigma}_1^- \hat{a}_1^\dagger), g_2 (\hat{\sigma}_2^+ \hat{a}_2 + \hat{\sigma}_2^- \hat{a}_2^\dagger)] \right. \\ \left. + [g_2 (\hat{\sigma}_2^+ \hat{a}_2 - \hat{\sigma}_2^- \hat{a}_2^\dagger), g_1 (\hat{\sigma}_1^+ \hat{a}_1 + \hat{\sigma}_1^- \hat{a}_1^\dagger)] + [g_2 (\hat{\sigma}_2^+ \hat{a}_2 - \hat{\sigma}_2^- \hat{a}_2^\dagger), g_2 (\hat{\sigma}_2^+ \hat{a}_2 + \hat{\sigma}_2^- \hat{a}_2^\dagger)] \right)$$

$$[g_1 (\hat{\sigma}_1^+ \hat{a}_1 - \hat{\sigma}_1^- \hat{a}_1^\dagger), g_1 (\hat{\sigma}_1^+ \hat{a}_1 + \hat{\sigma}_1^- \hat{a}_1^\dagger)] = g_1^2 ([\hat{\sigma}_1^+ \hat{a}_1, \hat{\sigma}_1^- \hat{a}_1^\dagger] - [\hat{\sigma}_1^- \hat{a}_1^\dagger, \hat{\sigma}_1^+ \hat{a}_1]) = g_1^2 ([\hat{\sigma}_1^+ \hat{a}_1, \hat{\sigma}_1^- \hat{a}_1^\dagger] + [\hat{\sigma}_1^- \hat{a}_1^\dagger, \hat{\sigma}_1^+ \hat{a}_1]) \\ = 2g_1^2 [\hat{a}_1 \hat{\sigma}_1^+, \hat{a}_1^\dagger \hat{\sigma}_1^-] = 2g_1^2 (\hat{a}_1 \hat{a}_1^\dagger [\hat{\sigma}_1^+, \hat{\sigma}_1^-] + [\hat{a}_1, \hat{a}_1^\dagger] \hat{\sigma}_1^+ \hat{\sigma}_1^-) = 2g_1^2 (\hat{a}_1 \hat{a}_1^\dagger + |e_1\rangle\langle e_1|)$$

$$\hat{\sigma}_1^+ \hat{\sigma}_1^- = |10\rangle\langle 10| + |00\rangle\langle 00| = |1\rangle\langle 1| \otimes |0\rangle\langle 0| \\ = -2g_1^2 [\hat{a}_1^\dagger \hat{\sigma}_1^-, \hat{a}_1 \hat{\sigma}_1^+] = -2g_1^2 ([\hat{a}_1^\dagger, \hat{\sigma}_1^-] \hat{\sigma}_1^+ + \hat{a}_1^\dagger [\hat{\sigma}_1^-, \hat{\sigma}_1^+]) = 2g_1^2 (\hat{\sigma}_1^- \hat{\sigma}_1^+ + \hat{a}_1^\dagger \hat{a}_1) \quad \text{(?)} \\ \text{analogously: } [g_2 (\hat{\sigma}_2^+ \hat{a}_2 - \hat{\sigma}_2^- \hat{a}_2^\dagger), g_2 (\hat{\sigma}_2^+ \hat{a}_2 + \hat{\sigma}_2^- \hat{a}_2^\dagger)] = 2g_2^2 (\hat{a}_2 \hat{a}_2^\dagger + |e_2\rangle\langle e_2|)$$

$$[g_i (\hat{\sigma}_i^+ \hat{a}_i - \hat{\sigma}_i^- \hat{a}_i^\dagger), g_j (\hat{\sigma}_j^+ \hat{a}_j + \hat{\sigma}_j^- \hat{a}_j^\dagger)]$$

$$= g_i g_j ([\hat{\sigma}_i^+ \hat{a}_i, \hat{\sigma}_j^+ \hat{a}_j] + [\hat{\sigma}_i^+ \hat{a}_i, \hat{\sigma}_j^- \hat{a}_j^\dagger] - [\hat{\sigma}_i^- \hat{a}_i^\dagger, \hat{\sigma}_j^+ \hat{a}_j] - [\hat{\sigma}_i^- \hat{a}_i^\dagger, \hat{\sigma}_j^- \hat{a}_j^\dagger])$$

$$= \hat{\sigma}_i^+ [\hat{a}_i, \hat{\sigma}_j^+ \hat{a}_j] + [\hat{\sigma}_i^+, \hat{\sigma}_j^+ \hat{a}_j] \hat{a}_i - [\hat{\sigma}_i^- \hat{a}_i^\dagger, \hat{\sigma}_j^+ \hat{a}_j] \\ = \hat{\sigma}_i^+ \hat{\sigma}_j^+ [\hat{a}_i, \hat{a}_j] + \hat{\sigma}_i^+ \hat{\sigma}_j^+ [\hat{a}_i, \hat{a}_j^\dagger] \hat{a}_j + [\hat{\sigma}_i^+, \hat{\sigma}_j^+] \hat{a}_i \hat{a}_j + \hat{\sigma}_i^- [\hat{a}_i^\dagger, \hat{\sigma}_j^+ \hat{a}_j] - [\hat{\sigma}_i^- \hat{a}_i^\dagger, \hat{\sigma}_j^+ \hat{a}_j] \\ = \hat{\sigma}_i^+ \hat{\sigma}_j^+ - \hat{\sigma}_i^- \hat{\sigma}_j^+ [\hat{a}_i^\dagger \hat{a}_j] \\ = \hat{\sigma}_i^+ \hat{\sigma}_j^+ + \hat{\sigma}_i^- \hat{\sigma}_j^+$$

$$\Rightarrow \text{all together: } \frac{i}{\omega_0} [S, H_g] = -\frac{\hbar}{\omega_0} \left(\sum_{i=1,2} 2g_i^2 (\hat{a}_i \hat{a}_i^\dagger + |e_i\rangle\langle e_i|) + 2g_1 g_2 (\hat{\sigma}_1^+ \hat{\sigma}_2^- + \hat{\sigma}_1^- \hat{\sigma}_2^+) \right)$$

$$\Rightarrow \tilde{H} = \sum_{i=1,2} \left[-\hbar \Delta |e\rangle\langle e| - \frac{\hbar g_i^2}{4\Delta} (\hat{a} \hat{a}^\dagger \sigma_i^z + |e\rangle\langle e|) \right] - \frac{\hbar g_1 g_2}{4\Delta} (\sigma_1^z \sigma_2^z + \sigma_1^x \sigma_2^x)$$

$$i\hbar \frac{\partial}{\partial t} |\tilde{\psi}\rangle = \tilde{H} |\tilde{\psi}\rangle$$

$$i\hbar \frac{\partial}{\partial t} U(t) |\tilde{\psi}\rangle = \tilde{H} U(t) |\tilde{\psi}\rangle$$

$$\frac{\partial}{\partial t} U(t) |\tilde{\psi}\rangle = \dot{U}(t) |\tilde{\psi}\rangle + U(t) \dot{|\tilde{\psi}\rangle} = \dot{U}(t) U^{-1}(t) |\tilde{\psi}\rangle + U(t) |\dot{|\tilde{\psi}\rangle}| = \tilde{H} U |\tilde{\psi}\rangle$$

$$\begin{aligned} \rightarrow U(t) |\tilde{\psi}\rangle &= \tilde{H} U |\tilde{\psi}\rangle - i\hbar \dot{U} U^{-1} |\tilde{\psi}\rangle \\ |\dot{|\tilde{\psi}\rangle}| &= U^{-1} \tilde{H} U |\tilde{\psi}\rangle - U^{-1} \dot{U} |\tilde{\psi}\rangle \end{aligned}$$

$$\Rightarrow \tilde{H} = U^{-1} H U - i\hbar \dot{U} U^{-1}$$

$$U = e^{-i\tilde{H}t/\hbar} = e^{-iHt/\hbar}$$

$$U^{-1} H U:$$

$$e^{-i\tilde{H}t/\hbar} \omega_g |e\rangle\langle e| e^{i\tilde{H}t/\hbar} = \omega_g |e\rangle\langle e|$$

$$e^{i\tilde{H}t/\hbar} \sum_n \omega_n \hat{a}^\dagger \hat{a} e^{-i\tilde{H}t/\hbar} = \sum_n \omega_n \hat{a}^\dagger \hat{a}$$

$$e^{i\tilde{H}t/\hbar} |n\rangle\langle n| e^{-i\tilde{H}t/\hbar} : e^{-i(\omega_g |e\rangle\langle e| + \sum_n \omega_n |n\rangle\langle n|)t} |n\rangle = e^{-i\omega_n t} |n\rangle$$

$$e^{i\tilde{H}t/\hbar} |e\rangle = e^{-i(\omega_g |e\rangle\langle e| + \sum_n \omega_n |n\rangle\langle n|)t} |e\rangle = e^{-i\omega_g t} |e\rangle$$

$$\Rightarrow e^{i\tilde{H}t/\hbar} |n\rangle\langle n| e^{-i\tilde{H}t/\hbar} = |n\rangle\langle n|$$

$$e^{i\tilde{H}t/\hbar} |e\rangle\langle n| e^{-i\tilde{H}t/\hbar} = |e\rangle\langle n|$$

$$\begin{aligned} \Rightarrow \tilde{H} &= \omega_g |e\rangle\langle e| + \sum_n \omega_n \hat{a}^\dagger \hat{a} + \frac{g}{\sqrt{2\pi M}} \sum_{n=1}^M |n\rangle\langle e| + |e\rangle\langle n| - \omega_g |e\rangle\langle e| - \omega_g \sum_n |n\rangle\langle n| \\ &= \sum_n (\omega_n - \omega_g) |n\rangle\langle n| + \frac{g}{\sqrt{2\pi M}} \sum_{n=1}^M |n\rangle\langle e| + |e\rangle\langle n| \end{aligned}$$

$$\hat{a}^\dagger \hat{a} = n |n\rangle\langle n|$$

$$b) |\psi_S(t)\rangle = e^{-i(H_{S0} + H_{S1})t}$$

$$|\tilde{\psi}_f(t)\rangle = e^{-iH_{S0}t/\hbar} e^{-iH_{S1}t/\hbar} |\psi_S(0)\rangle = e^{-iH_{S0}t/\hbar} |\psi_S(0)\rangle = e^{-iH_{S0}(t)} e^{iH_{S0}(t)} |\tilde{\psi}_f\rangle$$

$$|\tilde{\psi}_f(t)\rangle = e^{-i\tilde{H}t/\hbar} |\psi_S(0)\rangle$$

$$\sigma^+ \hat{a} + \sigma^- \hat{a}^\dagger$$

$$\tilde{H} |n+1, g\rangle = (n+1)(\omega_n - \omega_g) + \frac{g}{\sqrt{2\pi M}} \sqrt{n+1} |n, e\rangle$$

$$\tilde{H} |n, e\rangle = n(\omega_n - \omega_g) + \frac{g}{\sqrt{2\pi M}} \sqrt{n+1} |n+1, g\rangle$$

again just notes
I made
during the thought
process (just ignore them)

$$\sum_n (n\omega_n - \omega_{eg}) |n\rangle\langle n| + \frac{g}{\sqrt{2\pi\hbar}} \sum_{n=1}^M \sum_{k=1}^{B_n} |n\rangle\langle k| + |k\rangle\langle n|$$

Just notes I took while writing the code

$$\hat{F} = q_0|0\rangle + a_1|1\rangle + a_2|2\rangle + \dots + a_M|M\rangle, \hat{Q} = q_0|0\rangle + q_1|1\rangle$$

$\Rightarrow$ K-S. HR

$$A_n = (n\omega_n - \omega_{eg}) \begin{pmatrix} 0 & 0 & \dots & 1 & \dots & 0 \end{pmatrix} \otimes \mathbb{1} \quad |e\rangle|n\rangle|0\rangle$$

$$B_n: \hat{a}_{j,n}^\dagger = \frac{\sqrt{n}}{n} |n-1\rangle\langle n|$$

$$\hat{a}_{j,n} = \frac{1}{\sqrt{n}} |n\rangle\langle n-1|$$

$$B_n = \frac{g}{\sqrt{2\pi\hbar}} \left((\hat{a}_n^\dagger \otimes \sigma_-) + (\hat{a}_n \otimes \sigma_+) \right)$$

$$= \frac{g}{\sqrt{2\pi\hbar}} \left(\sqrt{n} |n\rangle\langle n-1| \otimes |0\rangle\langle 1| + \sqrt{n} |n-1\rangle\langle n| \otimes |1\rangle\langle 0| \right)$$

$$\omega_{eg} - B = \frac{2B}{\mu} \cdot n$$